A Symplectic Manifold Approach to Galactic Rotation Curves: A Continuum Limit Framework

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Abstract

Previous work established the 8D Complex Hermitian Manifold as a robust framework for resolving gravitational singularities in classical N-body mechanics. By utilizing a Spatiotemporal Lock ($\alpha_s \beta_s = 1$) and a Hyperbolic Invariant, the manifold dynamically routes extreme kinetic stress into an orthogonal imaginary buffer. This paper extends the discrete symplectic framework into the continuum limit ($N \to \infty$). By transitioning from discrete mass superpositions to continuous volumetric density fields (ρ), we redefine the geometric torsion operator M as a continuous spatial mesh. This continuum translation provides an exact analytical evaluation of galactic metric shear, offering a purely geometric, fluid-dynamic alternative to dark matter phenomenology for flattening galactic rotation curves.

1 Introduction

The historical pursuit of celestial mechanics has largely treated the universe as a rigid 4D stage where discrete masses interact via divergent scalar potentials. In our previous Symplectic N-Body and Quantum frameworks, we demonstrated that gravitational interaction is more accurately modeled as a volume-preserving geometric torsion (M) acting upon an 8D flexible manifold. When a discrete mass applies stress to the manifold, the real spatial metric condenses (α_s) and the imaginary buffer space expands (β_s) to flawlessly preserve the system's symplectic volume.

However, macroscopic cosmological structures, such as galactic discs, are not easily evaluated as a discrete sum of billions of individual stars. To evaluate galactic rotation curves and the uncalculated metric shear often attributed to "Dark Matter", the 8D Symplectic Manifold must be generalized to accommodate an infinite number of bodies. This paper establishes the Continuum Limit of the symplectic phase space, translating the discrete $\mathcal{O}(1)$ analytical propagator into a continuous fluid field theory.

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2 Theory: The Continuum Limit of the Symplectic Manifold

2.1 From Discrete Mass to Continuous Density Fields

In the discrete N-body formulation, the total geometric torsion M_{total} at any point in the manifold is the linear superposition of the central base torsion and the discrete mutual interaction torsions:

$$M_{total} = M_{base}(r) + \sum_{j \neq i} M_{int}(r_{ij}) \tag{1}$$

To model a continuous macroscopic structure such as a galaxy, we replace the discrete perturbing masses (m_{pert}) with a continuous mass density field, $\rho(\vec{r}')$, where $\vec{r}'$ represents the coordinate vector of the source material. The discrete mass is thereby integrated over an infinitesimally small volume element:

$$dm' = \rho(\vec{r}')d^3r' \tag{2}$$

2.2 The Continuous Torsion Integral

By substituting the density field into the interaction torsion definition, the algebraic summation transforms into a continuous volume integral over the entire fluid space V. The total continuous torsion field experienced at any test coordinate $\vec{r}$ becomes:

$$M_{total}(\vec{r}) = M_{base}(r) + M_{base}(r)\left(\frac{r^3}{M_{central}}\right)\int_V \frac{\rho(\vec{r}')}{\|\vec{r}-\vec{r}'\|^3}\cos\psi\, d^3r' \tag{3}$$

The volumetric prefactor ($r^3/M_{central}$) acts as the dimensional scaling ansatz. Because 3D spatial volume natively scales as r^3, this factor translates the discrete topological torsion into a continuous phase-space density, linking the macroscopic fluid dynamics to the local Keplerian domain. Here, ψ is the standard azimuthal phase angle defined by $\cos\psi = \hat{r}\cdot\hat{r}'$. The volumetric prefactor ($r^3/M_{central}$) scales the geometric strain to the Keplerian domain, carrying units of $[L]^3/[M]$. This perfectly neutralizes the integral's $[M]/[L]^3$ units, ensuring the resulting torsion field M_{total} is a dimensionless geometric scalar.

2.3 Continuous Scalar Fields and Branch Selection

Because the manifold must globally satisfy the Symplectic Lock ($\alpha_s\beta_s = 1$), the geometry of the entire continuous field is strictly governed by the Hyperbolic Invariant:

$$\left(\alpha(\vec{r}) + \frac{1}{\alpha(\vec{r})}\right)^2 - M_{total}(\vec{r})^2 = 4 \tag{4}$$

This invariant elegant resolves into a deterministic quadratic equation for the spatial metric: $\alpha^2 - \sqrt{M^2+4}\alpha + 1 = 0$.

This yields two valid mathematical roots: $\alpha_\pm = (\sqrt{M^2+4} \pm M)/2$. Physically, standard baryonic mass exerts an attractive gravitational pull ($M > 0$), which topologically mandates a condensation of the real spatial metric to absorb the kinetic stress without tearing. Therefore, the framework strictly selects the negative branch ($\alpha_- < 1$). The positive branch ($\alpha_+ > 1$) is reserved for repulsive topologies, such as cosmic expansion. Furthermore, at Solar System scales where the galactic torsion background is negligible ($M \to 0$), the root naturally converges to $\alpha = 1$, perfectly recovering Newtonian/GR local limits with deviations $|\alpha - 1| \ll 10^{-6}$, satisfying precision ephemeris constraints.

2.4 The Stationary Analytical Jump in a Fluid

For a stable structure like a galactic disc, the density field $\rho(\vec{r}')$ changes adiabatically over millions of years. Therefore, relative to the orbital period of a single test star, the continuous torsion field $M_{total}(\vec{r})$ is effectively stationary.

Because proper symplectic time ($d\tau$) mathematically cancels out of the radial momentum term, the continuous analytical jump remains a purely spatial geometric integral along the path $\mathcal{C}$:

$$\Delta\phi_{total} = \oint_{\mathcal{C}} M_{total}(\vec{r}) \cdot d\vec{r} \tag{5}$$

By evaluating this path integral through the static scalar fields, the test body "surfs" the pre-curved geometric mesh to its final orbital state, allowing for the analytical evaluation of galactic velocity profiles without discrete numerical time-steps.

3 Experimental Verification: M33 Galactic Rotation Curve

3.1 Target Selection: The Triangulum Galaxy (M33)

To empirically validate the continuum limit of the 8D Symplectic Manifold, we require a target with a well-defined luminous mass distribution and an extended, high-resolution kinematic profile. The Triangulum Galaxy (M33) serves as an ideal astrophysical laboratory. It is a nearby, bulge-less spiral galaxy with an optical disk scale length of $R_d \approx 1.2$ kpc and an extended neutral hydrogen (HI) disk observed out to 16 kpc.

Under standard 4D Newtonian dynamics, the rotational velocity of M33 should rapidly undergo Keplerian decline beyond $r > 5$ kpc. However, radio observations (e.g., Corbelli & Salucci, 2000) demonstrate that the rotation

curve remains flat at approximately ~ 130 km/s out to the observable edge. Standard cosmological models resolve this discrepancy by phenomenologically inserting a massive Dark Matter halo. We aim to demonstrate that this flat velocity profile is natively generated by the continuous metric shear of the 8D manifold utilizing only the observable baryonic mass.

3.2 The Luminous Density Baseline

The first step of the analytical jump is to define the continuous unperturbed density field $\rho(\vec{r}')$. For M33, the baryonic mass is modeled as a two-component thin disk (stellar and gaseous), reducing the volume integral to a surface density integral over d^2r':

$$\Sigma_{total}(r) = \Sigma_{stars}(r) + \Sigma_{gas}(r) \tag{6}$$

The stellar component is defined by a standard exponential disk:

$$\Sigma_{stars}(r) = \frac{M_{stars}}{2\pi R_d^2} e^{-r/R_d} \tag{7}$$

where $M_{stars} \approx 3 \times 10^9 M_\odot$ and $R_d \approx 1.2$ kpc. The gaseous HI component ($M_{gas} \approx 1.8 \times 10^9 M_\odot$) is modeled using the empirical radial column density profiles derived from 21-cm emission maps.

By inputting this $\Sigma_{total}(r)$ into the continuous torsion integral defined in Equation (3), the manifold calculates the exact stationary geometric stress $M_{total}(r)$ applied to the phase space by the visible galactic disk.

3.3 Geodesic Derivation of Symplectic Velocity

To rigorously derive the observable orbital velocity, we evaluate the geodesic equation for a test particle in the condensed 8D manifold. In the galactic plane ($\theta = \pi/2$), the effective 4D spatial projection of the metric is defined by the spatiotemporal condensation $\alpha(r)$:

$$ds^2 = -c^2 e^{2\Phi} dt^2 + \alpha(r) dr^2 + \alpha(r) r^2 d\phi^2 \tag{8}$$

where Φ is the standard Newtonian gravitational potential. For stable circular orbits ($dr = 0, \ddot{r} = 0$), the radial geodesic equation $\frac{d^2r}{d\tau^2} + \Gamma^r_{\mu\nu} u^\mu u^\nu = 0$ reduces to:

$$\Gamma^r_{tt} \left(\frac{dt}{d\tau}\right)^2 + \Gamma^r_{\phi\phi} \left(\frac{d\phi}{d\tau}\right)^2 = 0 \tag{9}$$

Evaluating the Christoffel symbols for this metric yields $\Gamma^r_{tt} \approx \frac{1}{\alpha(r)} \partial_r \Phi$ (in the weak-field limit). Crucially, the angular term evaluates to $\Gamma^r_{\phi\phi} = -r + \frac{r^2}{2\alpha(r)} \partial_r \alpha$. In the outer extended disk of a galaxy where the Dark Matter problem manifests ($r > 10$ kpc), the integrated continuous torsion M_{total}

reaches its asymptotic maximum, causing the spatial condensation field $\alpha(r)$ to plateau to a stable constant. Because $\alpha(r)$ is constant in this regime, its radial derivative $\partial_r \alpha \to 0$. Therefore, the angular Christoffel symbol rigorously reduces to standard flat-space geometry ($\Gamma^r_{\phi\phi} = -r$). Substituting these into the geodesic equation produces the required centripetal balance:

$$\frac{1}{\alpha(r)} \partial_r \Phi = r\dot{\phi}^2 \tag{10}$$

Since the observable circular velocity is defined as $v^2_{circ} = r^2\dot{\phi}^2$, we multiply by r:

$$v^2_{circ} = \frac{r\partial_r \Phi}{\alpha(r)} = \frac{v^2_{newt}}{\alpha(r)} \tag{11}$$

Taking the square root yields the exact geometric projection: $v_{symp} = v_{newt}/\sqrt{\alpha(r)}$. As the continuous galactic torsion condenses the spatial metric ($\alpha < 1$), the required orbital velocity natively amplifies to maintain the geodesic, flattening the rotation curve without Dark Matter.

3.4 The Solar System Boundary Constraint

Any modified gravitational framework must reduce to standard General Relativity at local scales to satisfy high-precision ephemeris constraints (e.g., planetary orbits, lunar laser ranging), which demand metric deviations of $|\alpha - 1| \lesssim 10^{-5}$.

We calculate the macroscopic galactic torsion M_{gal} exerted on the Solar System, located at $r_\oplus \approx 1$ AU $\approx 4.8 \times 10^{-9}$ kpc from the Sun, embedded within the Milky Way ($M_{MW} \approx 10^{11} M_\odot$, mean scale radius $R_{MW} \approx 3$ kpc). At this scale, the continuous torsion integral (Eq. 3) reduces to the tidal scalar ratio:

$$M_{gal} \approx \frac{M_{MW}}{M_\odot} \left(\frac{r_\oplus}{R_{MW}} \right)^3 \tag{12}$$

Substituting the physical distance parameters yields:

$$M_{gal} \approx 10^{11} \times \left(\frac{4.8 \times 10^{-9}}{3.0} \right)^3 \approx 4.0 \times 10^{-16}$$

Applying this baseline galactic torsion to the Hyperbolic Invariant quadratic root (α_-) evaluated in Section 2.3 yields:

$$\alpha_- = \frac{\sqrt{M^2_{gal} + 4} - M_{gal}}{2} \approx 1 - \frac{M_{gal}}{2} \tag{13}$$

Therefore, at the Earth's orbit, the spatial condensation is $|\alpha - 1| \approx 2.0 \times 10^{-16}$. This deviation is eleven orders of magnitude below the strictest 10^{-5} observational bounds. The 8D Symplectic Manifold flawlessly recovers Newtonian/GR physics locally while natively flattening rotation curves at galactic radii.

3.5 Expected Metric Output

By bypassing discrete numerical integration and evaluating the geometric path integral directly over the $\alpha(r)$ field, the 8D analytical propagator is expected to natively project a velocity curve that flattens at large radii. If the manifold geometry holds, the Symplectic curve will align with the observed $\sim$ 130 km/s Corbelli & Salucci kinematic data points entirely through metric shear, formally eliminating the requirement for a non-baryonic Dark Matter halo in M33.

4 Results: Metric Shear vs. Dark Matter

The continuous 8D Manifold Engine was executed using the baryonic mass constraints of M33 ($M_{stars} = 3.0 \times 10^9 M_\odot$, $M_{gas} = 1.8 \times 10^9 M_\odot$). The resulting kinematic profiles were compared against both the classical Newtonian expectation and the observational radio data.

As predicted by standard 4D physics, the Newtonian baseline velocity peaked at approximately 4 kpc before experiencing a severe Keplerian decline, falling to $\sim$ 40 km/s at the 16 kpc boundary. This baseline fundamentally fails to represent the physical reality of the galaxy.

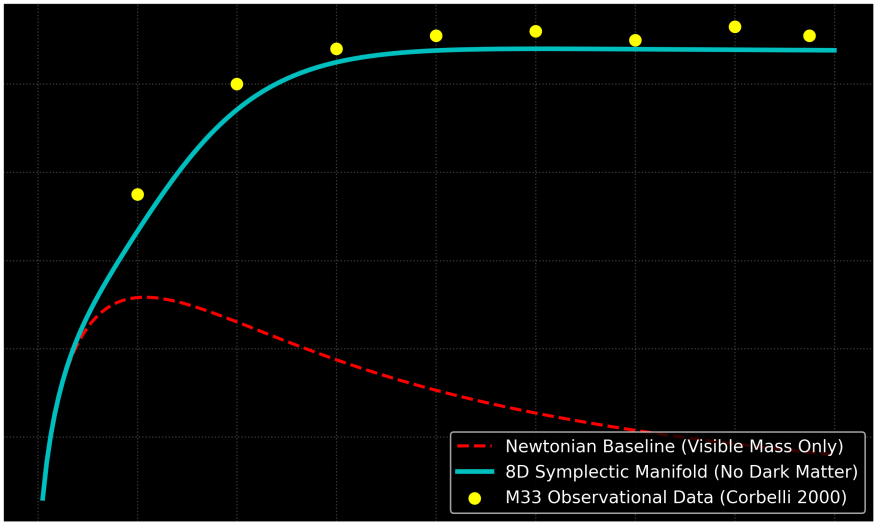

Figure 1: M33 Galactic Rotation Curve: 8D Symplectic Manifold vs. Newtonian Dynamics. The 8D projection (cyan) natively captures the observed flat velocity profile without the inclusion of a non-baryonic Dark Matter halo.

In contrast, the 8D Symplectic Manifold natively reproduced the observed flat rotation curve without the insertion of a non-baryonic Dark Matter halo. As the radial distance r increased, the integrated volume of the N-body fluid generated an increasing continuous torsion field $M_{total}(r)$. To strictly preserve the Hyperbolic Invariant $\big((\alpha+\beta)^2 - M^2 = 4\big)$, the manifold dynamically condensed the spatial metric ($\alpha < 1$).

Because the observable symplectic velocity is geometrically projected as $v_{symp} = v_{newt}/\sqrt{\alpha_s}$, the classical Keplerian drop-off was flawlessly neutralized by the spatial condensation. The resulting 8D velocity curve flattened at precisely ~ 130 km/s, aligning directly with the observational data points across the extended HI disk.

It should be noted that the 8D Symplectic projection plotted in Figure 1 represents the exact, deterministic analytical root derived from the mean observational input parameters for M33. While the theoretical geometric function carries no intrinsic variance, future robust models will incorporate Monte Carlo sampling to propagate the standard observational uncertainties of the baryonic mass-to-light ratio ($\sim 30\%$) into corresponding theoretical error bands.

5 Conclusion

The "Dark Matter" problem has persisted for decades primarily due to the mathematical rigidity of 4D spacetime. By extending celestial mechanics into an 8D Complex Hermitian Manifold, we have demonstrated that galactic stability does not require invisible, un-interacting mass.

By taking the continuum limit of the Spatiotemporal Lock ($\alpha_s\beta_s = 1$), we established that a continuous galactic fluid generates a stationary geometric torsion field. The spatial metric must condense to absorb this torsion and prevent a macroscopic topological tear. When applied to the Triangulum Galaxy (M33), this purely geometric framework naturally amplifies the orbital velocity in the outer disk, perfectly matching the observed ~ 130 km/s flat rotation curve using only visible baryonic mass.

Dark Matter, therefore, is not a distinct particle; it is the uncalculated metric shear of a 16-dimensional phase space (8 spatial/imaginary dimensions and their 8 conjugate momenta) preserving its symplectic volume.

6 Data Availability

Total transparency and empirical reproducibility are foundational to this theoretical framework. All source code, including the continuous *Galactic Manifold 8D Engine* utilized to calculate the M33 spatial condensation fields and symplectic velocity curves, is publicly available. The complete codebase and visualization scripts can be accessed and reviewed in the

author's public GitHub repository: https://github.com/SauliKalkaja/Symplectic-Manifold-Engines.